SAILESH IYYAPPAN . JR

Systems Product Engineer | Technical Founder

Embedded Systems • IoT • AI-Assisted Engineering • Research • Product Development

9486277916 | jr.saileshiyyappan@gmail.com | Erode, Tamil Nadu, India

Portfolio | LinkedIn | GitHub | GitLab

Professional Profile

Systems Product Engineer and Technical Founder building low-cost, end-to-end engineering products across embedded systems, IoT, AI-assisted software, and applied research. Experienced in turning ideas into deployable systems through research, architecture, hardware fabrication, firmware, software, testing, documentation, and technical validation.

10+ Projects • **8+ Hardware Prototypes** • **2+ Software Platforms** • **3+ Patent-Oriented Projects** • **Active Research**

Experience

Systems Product Engineer | Technical Founder

May 2019 – Present

Independent Engineering Projects | Erode, Tamil Nadu, India

- Execute the complete engineering cycle from idea research, architecture and blueprinting through fabrication, firmware, software, integration, testing, documentation and validation.
- Build resource-efficient prototypes using electronics, soldering, mechanical fabrication, drilling, CNC-oriented work, assembly, repair and AI-assisted development.

Selected Tier-1 Projects

GOATEST / GT-BAND

Jan 2024 – Present

Founder | Lead Engineer | Patent Application | Wearable Medical IoT

Modular wearable platform for physiological monitoring, personalized health assessment, AI-assisted decision support and emergency response. Integrates ESP32, physiological sensing, detachable hardware, BLE, firmware, mobile software and intelligent health-analysis workflows.

CRAFTLANCE

Feb 2026 – Present

Founder | Product Engineer | AI Engineering Startup Platform

AI-powered engineering collaboration platform connecting innovators, engineers, researchers and suppliers. Combines project intelligence, AI-assisted workflows, collaboration tools, cloud infrastructure and full-stack SaaS architecture.

HIGHLIGHTER BLUETOOTH SPEAKER

Mar 2023 – May 2023

Founder | Product Engineer | Patent-Oriented Consumer Electronics

Compact portable audio product built into a highlighter form factor, combining Bluetooth audio, standalone MP3 playback, MicroSD, rechargeable power, embedded electronics and mechanical design.

PAPER THIN SPEAKER – FCW

May 2025 – Present

Founder | Research Lead | Embedded Electronics / Acoustic Research

Research project developing an ultra-thin dynamic speaker using Flat Copper Winding (FCW), including a proof-of-concept 4-layer architecture for lightweight, low-profile acoustic systems.

ECO-INDUSTRIAL PARK (EIP)

Jul 2024 – Present

Founder | Research Lead | Industrial Systems / Sustainable Engineering

Systems research exploring an integrated circular recycling ecosystem through industrial symbiosis, automated dismantling, resource recovery, renewable energy and closed-loop material circulation.

Patents & Research

Patent Application Filed | Application No. 202641111694

17 Sep 2026

Modular Wearable System and Method for Personalized Physiological Monitoring, Longitudinal Health Assessment and Emergency Response

Published: 25 Sep 2026 | Status: Awaiting Request for Examination | Field: Bio-Medical Engineering

Patent Projects: GOATEST / GT-BAND • Highlighter Bluetooth Speaker

Research Projects: GOATEST / GT-BAND • CRAFTLANCE • Paper Thin Speaker – FCW • EIP

Literature review • Experimental development • Technical validation • Patent drawings • Research documentation • Publication preparation

Education

Integrated M.Tech – Computer Science and Engineering

Sep 2023 – Jun 2028

VIT-AP University, Amaravati, Andhra Pradesh, India

Embedded Systems • IoT • Artificial Intelligence • System Architecture • Applied Research • Product Development

Technical Stack

Embedded / IoT: ESP32 • ESP32-C3 • Arduino • PlatformIO • BLE • Wi-Fi • GSM • GPS • Sensors

Software / AI: C/C++ • Python • React • Next.js • TypeScript • Node.js • Firebase • Gemini API • Vertex AI • Machine Learning

Hardware / Fabrication: Electronics • Soldering • Mechanical Fabrication • Drilling • CNC • Assembly • Repair • Rapid Prototyping

Research / Tools: Git • GitHub • GitLab • Figma • draw.io • Patent Drawings • Technical Documentation • AI Prompting • Technical Validation

English – Fluent | **Tamil – Fluent** | Engineering & Product Innovation | Embedded Systems & IoT | Applied Research | AI-Assisted Engineering | Sustainable Engineering